

Sampling: Probability 101

```
$ echo "Data Sciences Institute"
```

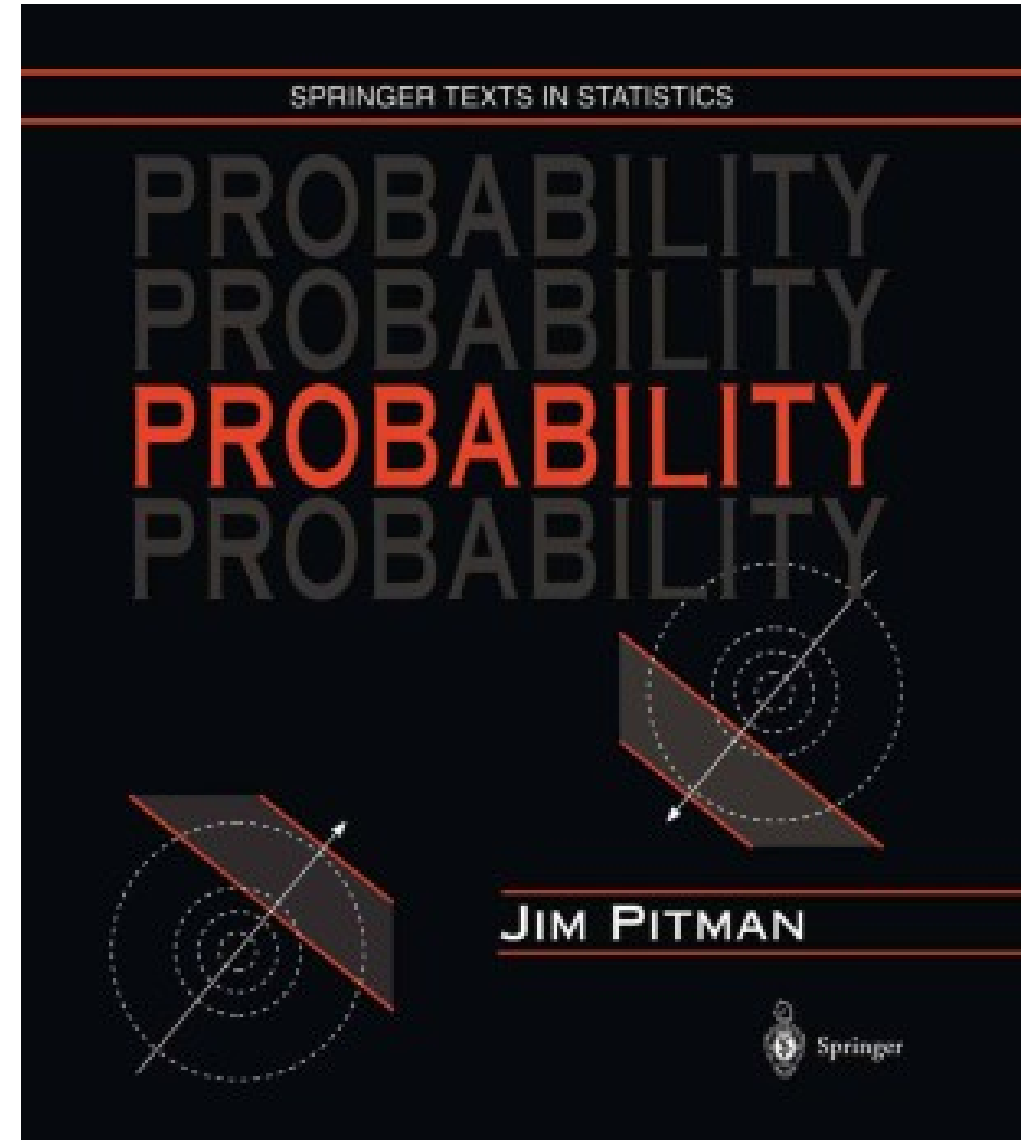
Learning Goals

- Introduce essential probability concepts
- Discuss statistical distributions (and why they matter!)
- Use examples to build statistical intuition

How do we calculate and interpret probabilities? What is a statistical distribution?

Key Texts

- Pitman, 1993, *Probability* , Springer, Chapters 1-3
- Image source: [Springer](#)



Before we start...

Intro to Probability

Definitions

- An **outcome space** is a set of all possible outcomes of some kind, often represented by Ω .
 - For example, $\Omega = \{A, B, C, \dots, Z\}$ is an outcome space containing all letters of the alphabet
- An **event** is a subset of an outcome space.
 - There are often many possible events for a specific outcome space
 - Possible events for Ω above could be vowels $\{A, E, I, O, U\}$ or letters before E $\{A, B, C, D\}$
- A **probability** is a function of an event describing how likely it is to occur

Equally Likely Outcomes

- If all outcomes in a set Ω are equally likely, the probability of event A is the number of outcomes in A divided by the total number of outcomes,


$$P(A) = \frac{\#A}{\#\Omega}$$

- $P(A)$ can be read as “the probability of A ”.

Example: Rolling a die

- For a six-sided die, the outcome space is,

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

- Some possible events and their probabilities are,

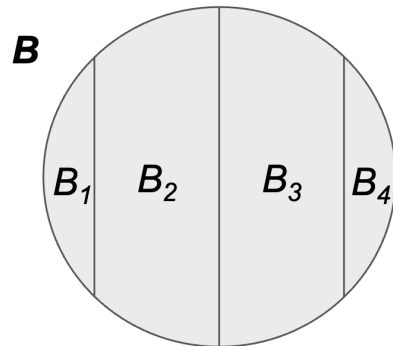
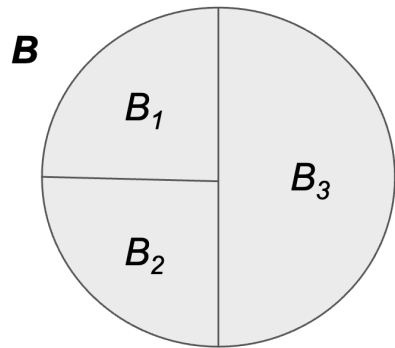
Description	Event	Probability
An even number is rolled	$A = \{2, 4, 6\}$	$3/6 = 1/2 = 0.5$
A number less than 6 is rolled	$B = \{1, 2, 3, 4, 5\}$	$5/6 = 0.833\dots$
A 6 is rolled	$C = \{6\}$	$1/6 = 0.166\dots$

Sets and Events

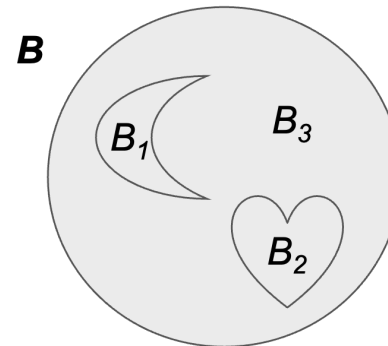
Event language	Set language	Set notation	Venn diagram
Outcome space	Universal set	Ω	
Event	Subset of Ω	A, B, C , etc.	
Impossible event	Empty set	$\emptyset$	
Not A , or the opposite of A	Complement of A	A^c	
Either A or B or both	Union of A and B	$A \cup B$	
Both A and B	Intersection of A and B	$A \cap B$	
A and B are mutually exclusive	A and B are disjoint	$A \cap B = \emptyset$	
If A , then B	A is a subset of B	$A \subseteq B$	

Partitions

- An event B is **partitioned** into n events $B_1, B_2, \dots, B_n$ if
 1. $B = B_1 \cup B_2 \cup \dots \cup B_n$ – every outcome in B belongs to some event B_i , none are left out
 2. $B_1, B_2, \dots, B_n$ are **mutually exclusive** – if an outcome is in event B_i , it is not in any other event



Adapted from Pitman (1993), Figure 1



Rules of Probability

- For an event B over an outcome space Ω ,
- **Non-negativity:** $P(B) \geq 0$
- **Addition:** If $B_1, B_2, \dots, B_n$ is a partition of B , then

$$P(B) = P(B_1) + P(B_2) + \dots + P(B_n)$$

- **Total one:** $P(\Omega) = 1$

Example: Drawing cards

- Suppose you have a regular deck of cards. Let B represent the event “drawing a heart”. Let B_1 and B_2 be a partition of B , with B_1 = “drawing non-numeric heart card (J, Q, K, A)” and B_2 = “drawing a numeric heart card (2,...,10)”
- B_1 and B_2 is a valid partition, since all heart cards are either numeric or non numeric, and a card cannot be both a numeric card and a non-numeric card (mutual exclusivity)

$$P(B) = 13/52 = 1/4 \geq 0$$

$$P(B_1) + P(B_2) = 4/52 + 9/52 = 13/52 = 1/4 = P(B)$$

$$P(\Omega) = P(\text{“draw any card in the deck”}) = 1$$

Conditional Probability

Conditional Probability

-  **Conditional probability** can be described as the probability that event A will happen **given that event B has already happened** 
- The notation for conditional probability is $P(A|B)$
- The formula for conditional probability is,

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$
- As long as $P(B) \neq 0$

Example: Rolling a die

- Suppose you roll a regular die, but haven't yet looked at the result. Let event A be "rolling a 4", and let event B be "rolling an even number".

- The probability that number you rolled is a 4 is,

$$P(A) = \frac{1}{6}$$

- Now suppose I look at the die and tell you that the number you rolled is **even** . Given this new information, the probability that the number you rolled is a 4 is,

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(\text{"rolling a 4" and "rolling an even number"})}{P(\text{"rolling an even number"})} = \frac{\frac{1}{6}}{\frac{1}{2}} = \frac{1}{3}$$

Independence

- If events A and B are **independent** , event A is not affected by the occurrence of event B.

- This can be described mathematically as,

$$P(A|B) = P(A)$$

- From the formula for conditional probability, we can then derive the following formula for independent events,

$$P(A \cap B) = P(A)P(B)$$

- The same applies for any number of disjoint events.

Random Variables

Random Variables

- **Random variables** are a way to describe a set of possible outcomes with a distribution of probabilities over the set of outcomes.
- Usually denoted with capital letters: X , Y , Z , etc.
- Random variables are similar to events
 - Events are a specific outcome or set of outcomes, while random variables describe possible outcomes and their various probabilities
 - An event "Number on the die is a five" or $\{5\}$ or $X = 5$ is one possible outcome of the random variable X , "the number obtained by rolling a die"

Random Variables

- The **range** of a random variable is all of the values it could possibly take. This can be continuous ($0 \leq X \leq 10$) or discrete ($X \in \{1, 2, 3\}$).
- The distribution of random variable is determined by the probabilities of values within its range,

$$P(X = x) \text{ for } x \in \text{range of } X$$

Example: Rolling two dice

- Let X represent the sum of the faces showing on two rolled dice.

x	2	3	4	5	6	7	8	9	10	11	12
Possible outcomes	1+1	1+2, 2+1	1+3, 2+2, 3+1	1+4, 2+3, 3+2, 4+1	1+5, 2+4, 3+3, 4+2, 5+1	1+6, 2+5, 3+4, 4+3, 5+2, 6+1	2+6, 3+5, 4+4, 5+3, 6+2	3+6, 4+5, 5+4, 6+3	4+6, 5+5, 6+4	5+6, 6+5	6+6
$P(X = x)$	1/36	1/18	1/12	1/9	5/36	1/6	5/36	1/9	1/12	1/18	1/36

Indicator Variables

- **Indicator variables** , denoted I_A , are a specific type of random variable that take the value 0 or 1 to indicate the occurrence of a given event A .
- Some examples of indicator variables may be votes in a two-party election (with event A being a vote for a particular candidate), votes for or against a bill, satisfied versus not satisfied reviews for a product, etc.

Distributions

Distributions

- ⚠️ A **probability distribution** is a statistical function that describes the probabilities of all possible events in an outcome space. ⚠️
- Distributions can be **discrete** (if the outcome space is distinct events, like rolling a die) or **continuous** (if the outcome space is a range of values, like choosing any real number between 1 and 10).
- Some features of interest for distributions may be their mean, variance, mode, skew, etc.

Binomial Distribution

- The **binomial distribution** concerns sequences of events with two possible outcomes: **success** and **failure**.
- Success occurs with probability p , and failure occurs with probability $q = 1 - p$. Trials defined this way are called **Bernoulli trials** .
- ⚠ The binomial distribution helps determine the probability of getting k **successes** in n **independent trials (with replacement)** ⚠

Binomial Distribution Formula

- For n independent trials with probability p of success and probability $q = 1 - p$ of failure, the probability of **k success in n independent trials** (with replacement) is,

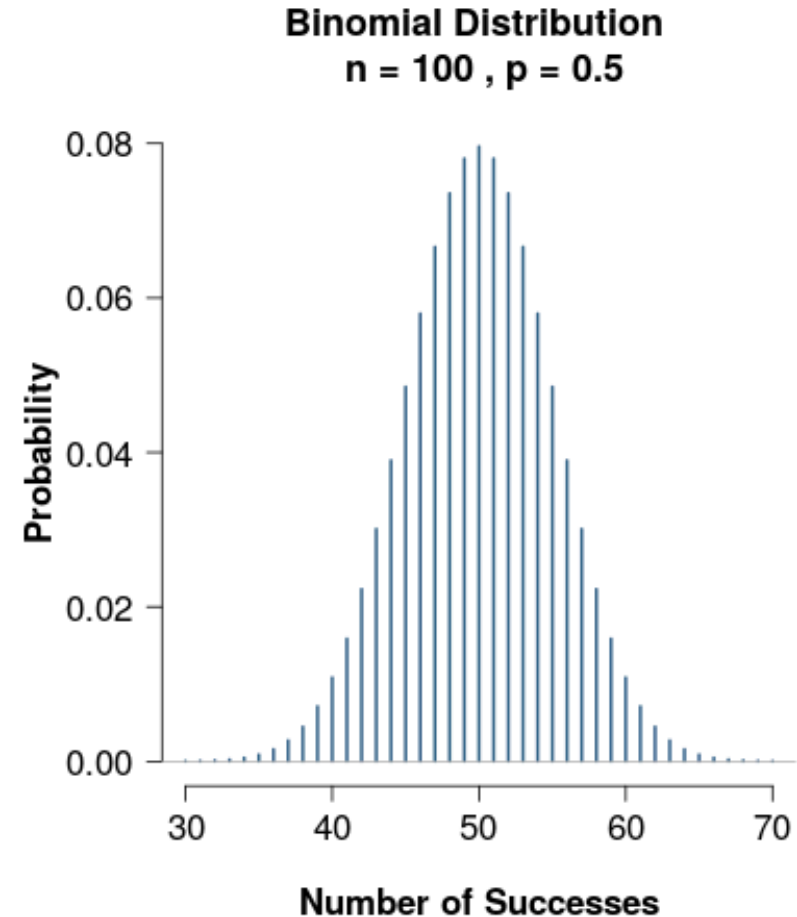
$$P(k \text{ success in } n \text{ trials}) = \binom{n}{k} p^k q^{n-k}$$

- $\binom{n}{k}$ is called " n choose k " and describes the number of possible combinations of k successes and $n - k$ failures:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Binomial Distribution

- For a large number of trials n , we expect the number of successes to be about np .
- For $n = 100$ and $p = 0.5$, the distribution of number of successes is centered around 50 (the most likely) and the total number of successes gets less likely as the numbers get farther from 50.



Example: Drawing cards

- Suppose you draw $n = 5$ cards from a standard deck, and your desired outcome is drawing a club. Then $p = 13/52 = 1/4$ and $q = 1 - p = 1 - 1/4 = 3/4$.
- For $k = 1$ success, the possible combinations of cards drawn are:

$\{\rightarrow \clubsuit \leftarrow, 2, 3, 4, 5\},$
 $\{1, \rightarrow \clubsuit \leftarrow, 3, 4, 5\},$
 $\{1, 2, \rightarrow \clubsuit \leftarrow, 4, 5\},$
 $\{1, 2, 3, \rightarrow \clubsuit \leftarrow, 5\},$
 $\{1, 2, 3, 4, \rightarrow \clubsuit \leftarrow\}$

- Mathematically this can be represented as,

$$\binom{5}{1} = \frac{5!}{1!(5-1)!} = \frac{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{1(4 \cdot 3 \cdot 2 \cdot 1)} = 5$$

Example: Drawing cards

- Since the trials are independent, the probability of getting 1 success in 5 trials is the product of the probability of getting a club on one trial and the probability of getting non-clubs on four trials.

$$P(\text{Club})P(\text{Non-club})P(\text{Non-club})P(\text{Non-club})P(\text{Non-club}) = \frac{1}{4} \cdot \frac{3}{4} \cdot \frac{3}{4} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{1^1}{4} \cdot \frac{3^4}{4} = 0.0791$$

- Putting the two calculations together, we have,

$$P(1 \text{ club in 5 draws}) = \binom{5}{1} \frac{1^1}{4} \cdot \frac{3^4}{4} = 5 \cdot 0.0791 = 0.396$$

Uniform Distribution

- The **uniform distribution** describes a situation in which every outcome on a certain set or interval is **equally likely** .
- This can be represented mathematically as,

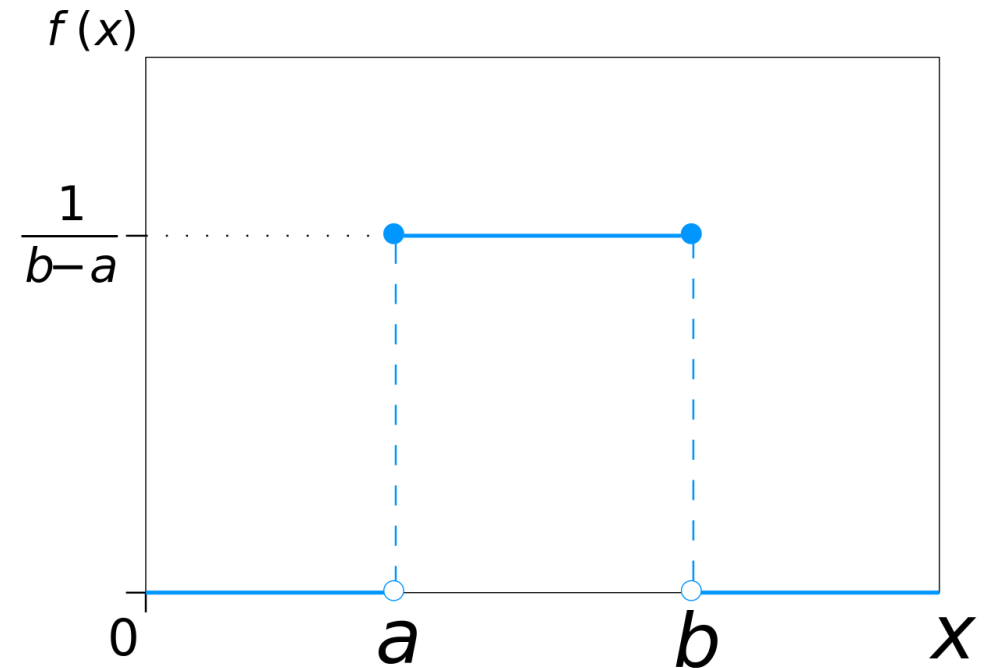

$$P(X = x) = \begin{cases} \frac{1}{|\Omega|} & \text{if } x \in \Omega, \\ 0 & \text{otherwise} \end{cases}$$

or


$$P(X = x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b, \\ 0 & \text{otherwise} \end{cases}$$

Uniform Distribution

- Examples
 - Rolling a die
 - Drawing any card from a normal deck
 - Choosing a random number between 1 and 100
 - Choosing a random student in a classroom



Poisson Distribution

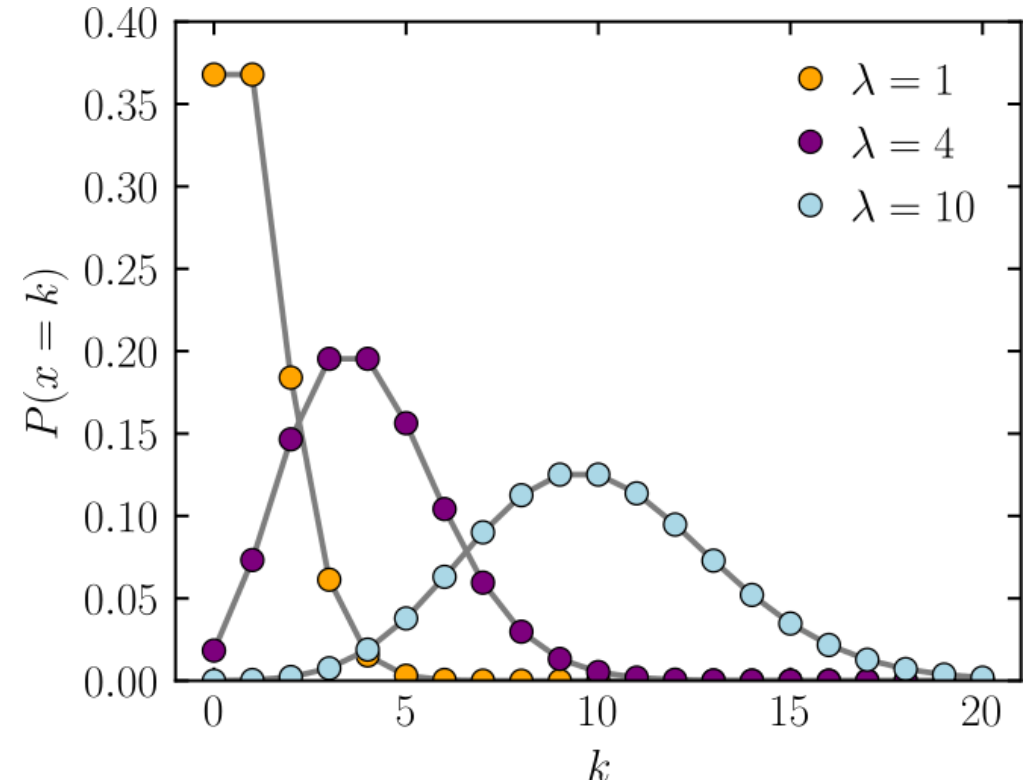
- The **Poisson distribution** is an approximation of the distribution of the **number N of occurrences of events of some kind**, when the events all have small probabilities and are independent.

$$P(N = k) \approx \frac{e^{-\mu} \mu^k}{k!} \text{ for } k = 1, 2, \dots$$

- The Poisson distribution is a discrete probability distribution.

Poisson Distribution

- Examples
 - Number of wins in n games of roulette for a gambler who bets on a single number each game
 - Number of rain drops that land on a particular area of a roof during a set time interval
 - Number of people who enter a store in a certain time interval



Normal Distribution

- The **normal distribution** is one of the most common and important distributions
- It is represented by the equation,

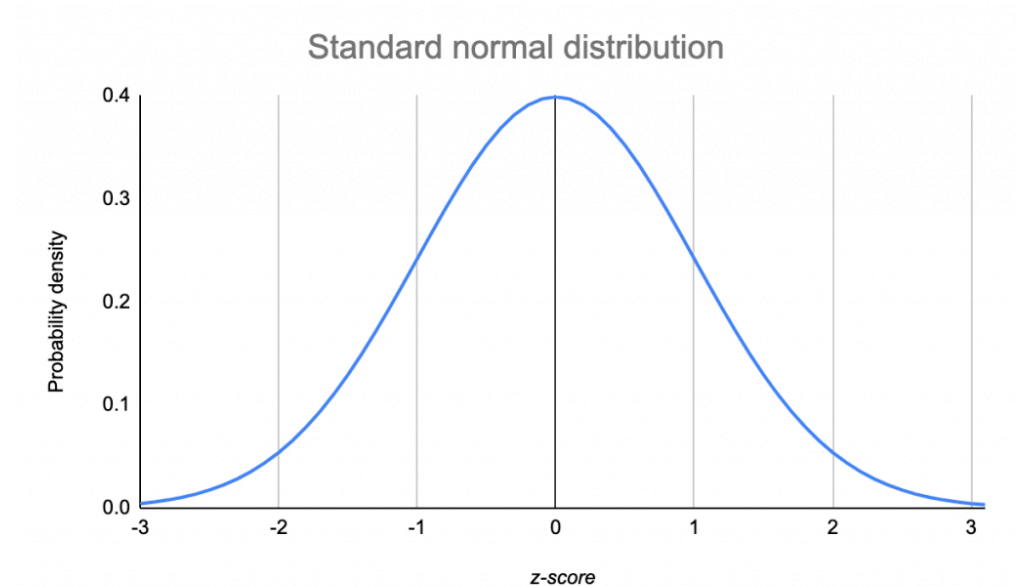
$$P(X = x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- where μ is the mean of X and σ is the standard deviation.
- A random variable X following a normal distribution is often denoted,

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

Normal Distribution

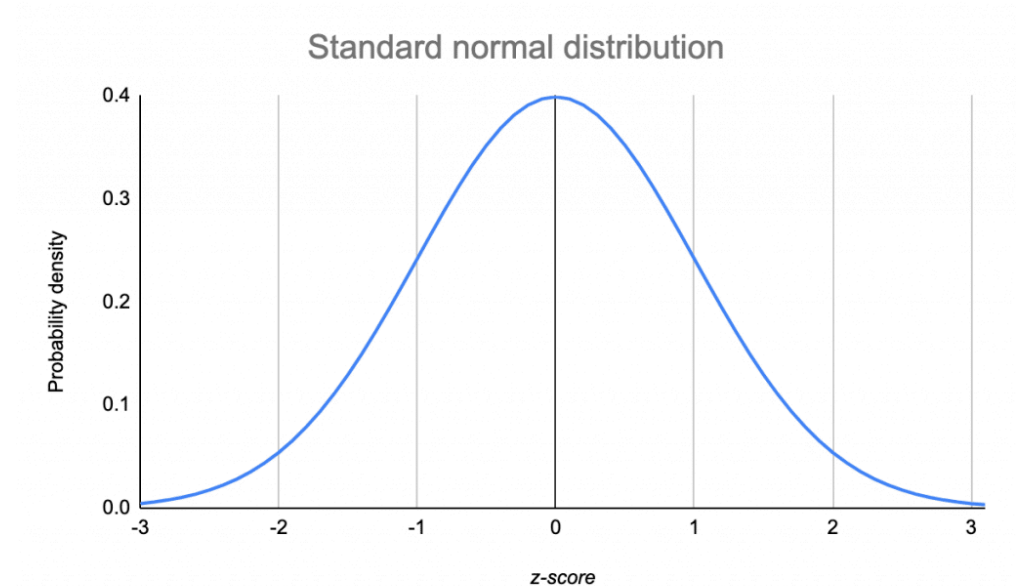
- The normal distribution is centered and symmetric about μ . σ describes the horizontal spread (how wide the distribution is).
- The normal distribution can be used to approximate other distributions for easy calculations of probabilities



Standard Normal Distribution

- The standard normal distribution is a normal distribution with mean 0 and standard deviation 1.
- In general, a random variable X with a normal distribution can be standardized using the following formula,

$$Z = \frac{X - \mu}{\sigma}$$



Example: Z-Scores

- Z-Scores represent the probability that a value is less than or equal to the value of a given standardized random variable.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

Expected Value

Expected Value

- The **expected value** or **expectation** of a random variable X is the mean of the distribution of X , denoted $E(X)$ or μ . This is represented mathematically as,

$$E(X) = \sum_{\text{Every } x} xP(X = x)$$

- The expected value is the average of all possible values of X weighted by their probabilities.
- The expected value of indicator variable I_A is the probability of event A ,

$$E(I_A) = P(A)$$

Example: Sampling a student

- Suppose you are randomly sampling a student from a school. There are 200 students each of ages 16, 17, and 18. Let random variable X represent the age of the student sampled.
- The expected age of the student selected is,

$$E(X) = \sum_{\text{Every } x} xP(X = x) = 16\left(\frac{200}{600}\right) + 17\left(\frac{200}{600}\right) + 18\left(\frac{200}{600}\right) = 17$$

- Now suppose there are 100 students age 16, 200 students age 17, and 300 students age 18. The new expected age is,

$$E(X) = \sum_{\text{Every } x} xP(X = x) = 16\left(\frac{100}{600}\right) + 17\left(\frac{200}{600}\right) + 18\left(\frac{300}{600}\right) = 17.333$$

Properties of Expectation

- Constants : The expectation of a constant random variable is its constant value

$$E(c) = c$$

- Scalar multiplication : For random variable X multiplied by constant c ,

$$E(cX) = cE(X)$$

- Addition : The expectation of a sum of random variables is the sum of the expectations

$$E(X + Y) = E(X) + E(Y)$$

Variance and Standard Deviation

Variance and Standard Deviation

- The **variance** of X , denoted $\text{Var}(X)$, is the mean squared deviation of X from its expected value $E(X)$,

$$\text{Var}(x) = E([X - E(X)]^2) = E(X^2) - [E(X)]^2$$

- The **standard deviation** of X , denoted $\text{SD}(X)$, is the square root of the variance of X :

$$\text{SD}(X) = \sqrt{\text{Var}(X)}$$

Variance Properties

- **Addition:** for independent random variables $X_1, X_2, \dots, X_n$, the variance of their sum is,

$$Var(X_1 + \dots + X_n) = Var(X_1) + \dots + Var(X_n)$$

- **Scalar multiplication:** for random variable X and scalar c ,

$$Var(cX) = c^2 Var(X)$$

Variance and Standard Deviation

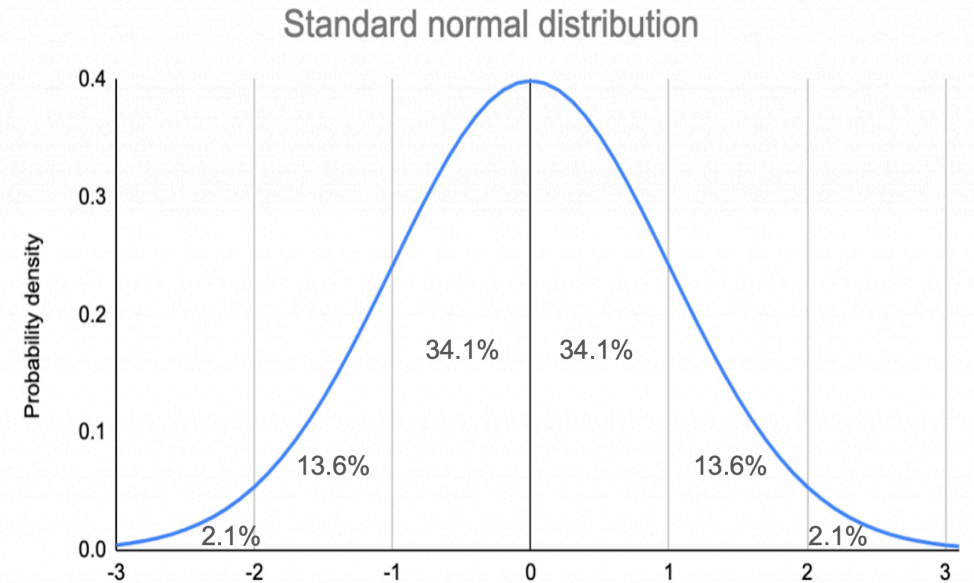
- Variance is often denoted σ^2 , with SD denoted σ
- Variance and SD describe how spread out the distribution of a variable is
- SD is often easier to interpret since its units are the same as the mean
-  In general 
 - For a random variable X with some distribution, you should expect the value of X to be around the expected value $E(X)$, plus or minus a few times the standard deviation $SD(X)$

Example: Normal Distribution

- On a normal distribution, ~68% of the probability density lies within one SD of the mean:

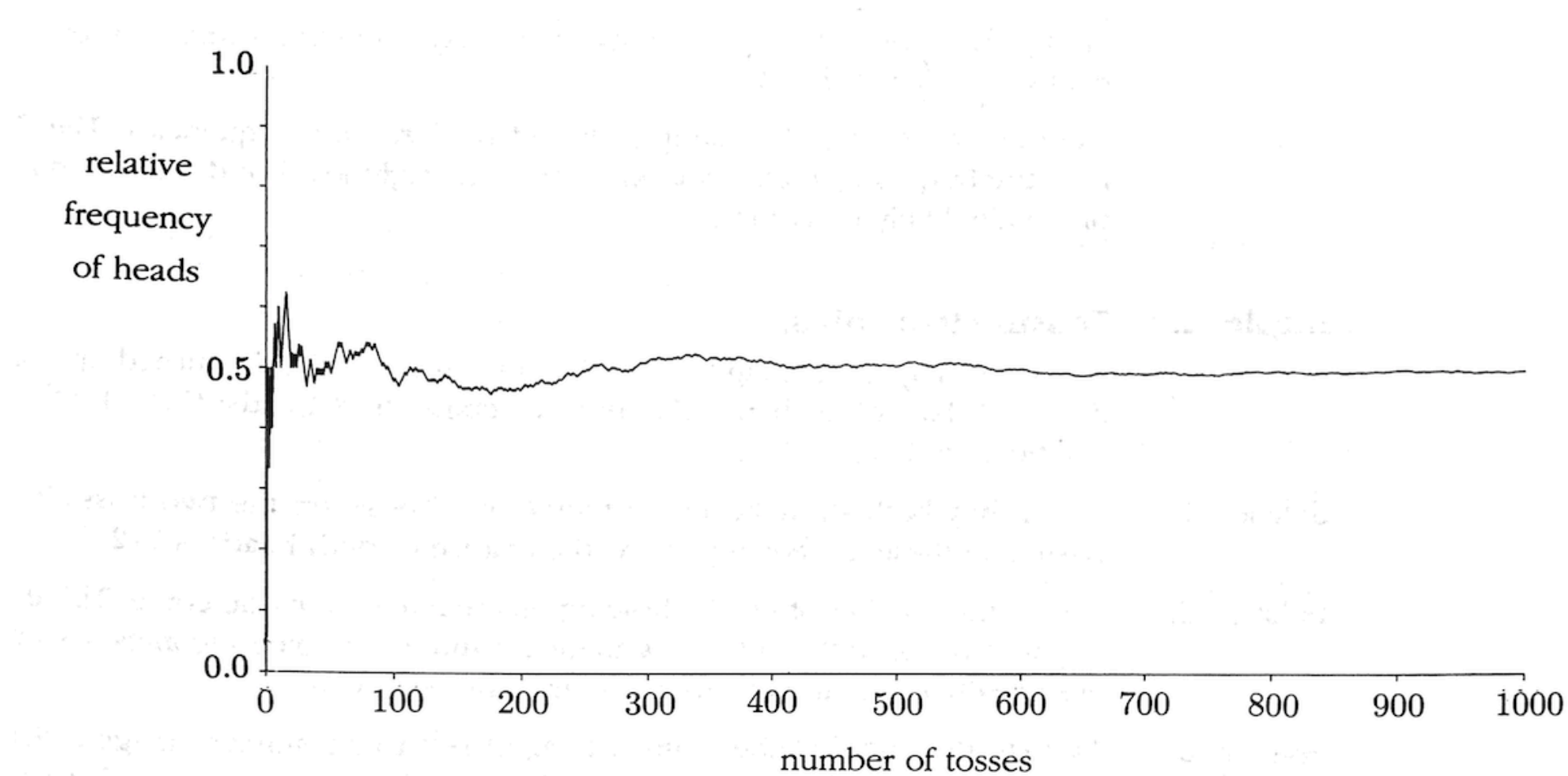
■ $P(\mu - \sigma \leq X \leq \mu + \sigma) \approx 0.68$

- For the standard normal $N(0, 1)$, this means X is fairly likely to be between -1 and 1, and that 2.5 would be a very unlikely value of X .



Law of Large Numbers

Law of Large Numbers



Law of Large Numbers

- ⚠ If the number of trials n is large, the proportion of successes in n independent trials will, with overwhelming probability, be very close to p , the probability of success on each trial ⚠
- **Intuition for sampling:**
 - As the number of units sampled increases, the proportion of units that exhibits a certain trait will grow closer and closer to the true proportion of individuals in the population with that trait

Central Limit Theorem

Central Limit Theorem

- ⚠ Let $S_n = X_1 + X_2 + \dots + X_n$ be the sum of n independent random variables each with the same distribution. For large n , the distribution of S_n is approximately normal, with mean $E(S_n) = n\mu$ and variance $\text{Var}(S_n) = \sigma^2_n$, where $\mu = E(X_i)$ and $\sigma^2 = \text{Var}(X_i)$. ⚠
- This also holds true for,

$$\bar{X}_n = \frac{S_n}{n}, \text{ with } E(\bar{X}_n) = \mu \text{ and } \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$$

$$\bar{X}_n \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

Who cares?

- Sample size and quantity affect how well our conclusions can represent our target population (LLN, CLT)
- Distributions and their assumptions affect our models and the ways that we calculate various statistics



Next

- Probability vs Non-probability sampling
- Sampling Types and Sample Design
- Observational Studies and Censuses